

Lab M12.03 – Scheduling and Batch Strategy Design (Problem Lab)

Part 1: Cron Expression Exercises

- | | |
|--------------------------|--------------|
| 1. Daily at 2 AM | 0 2 * * * |
| 2. Hourly at :15 | 15 * * * * |
| 3. Monday 6 AM | 0 6 * * 1 |
| 4. 1st of month midnight | 0 0 1 * * |
| 5. Every 15 min | */15 * * * * |
| 6. Weekdays 8 AM | 0 8 * * 1-5 |

Part 2: Pipeline Scheduling Design

Pipeline 1: Nightly Warehouse Refresh

- Scheduling Strategy: Time-based
- Schedule/Trigger:
 - If time-based: 04***
- Justification:
 1. Why this strategy? Because source database backup completes around 1 AM and there is a time limit for the BI dashboards (6AM). Therefore, it would be safe to schedule any time in between (4AM).
 2. Why not the alternatives? it could also be dependency-aware and be triggered after source database backup completes.
- Failure handling:
 - What if it fails? Re-run, alert if not completed by 5AM
 - Is it safe to re-run? use overwrite or incremental strategy

Pipeline 2: Hourly Clickstream Aggregation

- Scheduling Strategy: Event-based
- Schedule/Trigger:
 - If event-based: Trigger when new files arrive
- Justification:
 1. Why this strategy? Event-based ensures all data is present before processing
 2. Why not the alternatives? Not time-based: risk of partial data. Not dependency-aware: no strict upstream pipeline

- Failure handling:
 - What if it fails? Track processed files with checkpoints
 - Is it safe to re-run? using partition overwrite

Pipeline 3: Financial Close Pipeline

- Scheduling Strategy: Dependency-aware
- Schedule/Trigger:
 - If dependency-aware: each step depends on previous (extract → validate → calculate → report → email)
- Justification:
 1. Why this strategy? Multi-step workflow
 2. Why not the alternatives? Because it needs orchestration between steps
- Failure handling:
 - What if it fails? Maintain intermediate state checkpoints and flags after each step (validation)
 - Is it safe to re-run? recalculations overwrite previous results. Re-run is time heavy

Pipeline 4: Partner File Ingestion

- Scheduling Strategy: Event-based
- Schedule/Trigger:
 - If event-based: Trigger when file appears on SFTP
- Justification:
 1. Why this strategy? Because file arrival is unpredictable
 2. Why not the alternatives? Not dependency-aware because there is no upstream pipeline, just file existence
- Failure handling:
 - What if it fails? Separate processed files to avoid duplicates
 - Is it safe to re-run? Full overwrite as files are small

Pipeline 5: ML Feature Pipeline

- Scheduling Strategy: Dependency-aware
- Schedule/Trigger:
 - If dependency-aware: Trigger after Pipeline 1 completes successfully

- Justification:
 1. Why this strategy? Strong dependency on fresh warehouse data
 2. Why not the alternatives? Not time-based because it could run before data is ready
- Failure handling:
 - What if it fails? Alert if not complete by 6:30 AM
 - Is it safe to re-run? overwrite

Pipeline 6: Data Quality Checks

- Scheduling Strategy: Dependency-aware
- Schedule/Trigger:
 - If dependency-aware: Trigger after each upstream pipeline completes
- Justification:
 1. Why this strategy? Must validate outputs immediately after each pipeline
 2. Why not the alternatives? Not time-based, unpredictable timing, could miss failures or run too late.
- Failure handling:
 - What if it fails? Alert on failure and audit results
 - Is it safe to re-run? Yes, just queries for validation

Part 3: Anti-Pattern Identification

Setup 1

Pipeline A: Extract data from API	→ Cron: 0 1 * * * (1:00 AM)
Pipeline B: Transform extracted data	→ Cron: 0 2 * * * (2:00 AM)
Pipeline C: Load transformed data	→ Cron: 0 3 * * * (3:00 AM)

It should be dependency-aware workflow. Must trigger B after A succeeds and trigger C after B succeeds in order to guarantee correct execution order .

Setup 2

Pipeline: Load partner file
Schedule: Every 5 minutes, check if file exists, process if found
Problem: File is uploaded once per week

It does not make sense to schedule every 5 minutes if the file is uploaded once per week. It should be event-based and trigger when the file arrives

Setup 3

Pipeline: Nightly warehouse refresh

Schedule: 0 0 * * * (midnight)

Note: Source database backup runs from 11 PM to 1 AM

Pipeline may read incomplete data. Adjust cron to safe time: 02***.